Domain Adaptive Cross Reconstruction for Change Detection of Heterogeneous Remote Sensing Images via a Feedback Guidance Mechanism

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***Abstract*— Change detection on heterogeneous optical and synthetic aperture radar (SAR) images is soaring and plays a crucial role in monitoring land cover changes, such as disaster emergencies and natural resource monitoring. This is commonly recognized as a promising but challenging work due to the intrinsic differences in imaging mechanisms between the optical and SAR images. Recently, deep learning (DL)-based change detection methods based on two-step processing have attracted attention, i.e., first image translation between optical and SAR images to alleviate their modality differences and then change detection based on the translated images. However, image translation itself is a trouble task for the heterogeneous optical and SAR images. The unreliable image translation results further limit the accuracy of change detection. In this article, to mitigate this problem, we propose a change detection model on domain adaptation by novelty integrating change detection and image reconstruction into a unified framework. Specifically, we first transform the optical and SAR images into an intermediate common domain for comparison. Moreover, cross reconstruction for optical and SAR images is designed to maintain the characteristics of the images and improve the performance of domain adaptation. In addition, a feedback guidance mechanism is circumspectly designed to co-optimize change detection and image reconstruction tasks. Extensive experiments were conducted on four publicly available datasets, and the results demonstrate the effectiveness of our proposed method.**

***Index Terms*— Change detection, cross reconstruction, domain adaptation, feedback guidance mechanism, remote sensing.**

I. Introduction

REMOTE sensing change detection has a wide range of practical applications in monitoring the Earth's surface, such as urban planning [1], [2], disaster assessment [3], [4], and resource monitoring [5], [6]. Generally, change detection aims to identify the land cover changes between the multitemporal images in the same geographical area [7], [8], [9]. Since optical remote sensing images can provide spectral information, texture, and geometry of ground features, the change detection in optical images has received more attention [10], [11], [12], [13]. However, optical sensors are inevitably affected by cloud and weather conditions, so, in some cases, it is difficult to obtain effective multitemporal optical images seasonable in the same region [14]. In contrast, synthetic aperture radar (SAR) images have all-time and all-weather advantages, which can provide effective and reliable images in severe weather conditions [15]. These complementary characters between optical and SAR images make them indispensable in disaster assessment, such as a flood or fire. Therefore, it is more in line with reality and meaningful to investigate the change detection in heterogeneous optical and SAR images.

Traditional change detection methods for heterogeneous images always employ similarity measures or regression approaches to determine the change regions. Unfortunately, it is difficult to identify the relationship between the heterogeneous images when the scene is complex and noise in SAR images is severe. Recently, deep learning (DL)-based methods have attracted plenty of attention due to their amazing performance. Most existing heterogeneous image change detection methods always have three steps [16], [17], [18], [19]. First, they translate one image domain to another image domain via a translation network. Second, the difference image (DI) is obtained through direct comparison or other strategies. Finally, the change regions are obtained by using threshold segmentation, for example, an unsupervised change detection (USCD) [16] model for heterogeneous images, which transformed one image space (e.g., optical image) to another image (e.g., SAR image) for comparison. A deep translation network for change detection (DTCDN) [17] is proposed, which performed image transformation through a deep transformation network to reduce the modality difference and then employed a supervised change detection network to determine the change regions.

Although DL-based methods have achieved desirable performance, several problems have not been solved yet. Specifically, most DL-based methods always translate an image domain to another image domain to decrease the modality difference based on generative adversarial network (GAN) [20] or its variants. For example, USCD employed a cycle-consistent adversarial network (CycleGAN) [21]

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to model the mapping relationship between heterogeneous images and DTCDN employed a no-independent-component-for-encoding GAN (NICE-GAN) [22] as the translation network. Although the translation network can alleviate the modality difference between heterogeneous images to a certain extent, it is generally difficult to achieve a high translation accuracy due to the district difference in imaging mechanisms and statistical properties between the two images. Correspondingly, the change detection result on the translated image is generally far from satisfactory. In conclusion, the existing change detection methods for heterogeneous images still have some major challenges; on one hand, numerous methods alleviate the difference in heterogeneous images through image translation [i.e., translating images from one domain (e.g., optical) to another domain (e.g., SAR)], which brings about the problem of error propagation. On the other hand, existing methods focus more on homogeneous information mining of the same modality after image translation, and they ignore the complementary information between the original heterogeneous images.

In this article, we propose a domain adaptation-based change detection for heterogeneous remote sensing images (DAHCD). Specifically, to mitigate the modality differences between the heterogeneous images, we first encode the two images to the intermediate common domain through domain adaptation, in which collaborative adaptations from both feature adaptation and cross reconstruction are employed. Feature adaptation enables heterogeneous images to be encoded into the same feature domain from the perspective of feature distribution. Cross reconstruction aims to reconstruct the SAR and optical images using the features of optical and SAR images, respectively. In this collaborative adaptation fashion, the features of optical and SAR images can be translated to an intermediate common domain. Moreover, a change detection network is proposed by considering the transformed multihierarchical features of optical and SAR images and the original heterogeneous optical-SAR image pairs, which consist of a multihierarchical difference information learning module and a difference information guidance module (DIGM). In addition, a feedback guidance mechanism is designed to utilize the result of change detection to guide the cross reconstruction for eliminating the influence of changing regions on the domain adaptive cross reconstruction. Finally, we design a weight decay strategy to dynamically adjust the weight of the loss function so that the network can optimize different parts in different training stages.

The main contributions of our proposed model are summarized as follows.

1) We propose an interesting domain adaptation-based change detection method for heterogeneous remote sensing images (DAHCD), which is the first time to eliminate the differences in heterogeneous images by integrating change detection and image reconstruction into a unified framework.
2) We propose a change detection module by comprehensively exploiting the intramodal information and intermodal information in the transformed multihierarchical features of optical and SAR images and the original heterogeneous optical-SAR image pairs, respectively. In addition, a feedback guidance mechanism is innovatively designed to collaborate representing change detection and cross reconstruction.

The rest of this article is organized as follows. The related works are listed in Section II. Section III illustrates the detailed framework of our proposed DAHCD. The details and analysis of the experiment are shown in Section IV. Section V gives the discussions of our proposed DAHCD. Finally, the conclusions are summarized in Section VI.

II. Related Works

A. Change Detection for Homogeneous Remote Sensing Images

Change detection on homogeneous images, such as bitemporal optical images with consistent feature representation, has been extensively investigated [9], [23], which can be roughly divided into two categories: traditional methods and DL-based methods.

Traditional methods can be subdivided into pixel-based methods and object-based methods. The pixel-based methods always calculate the DI through algebraic operations and then classify each pixel to obtain change regions [24]. The commonly used algebraic operations are the ratio method [25], the change vector analysis method [26], and the principal component analysis (PCA) method [27]. The results of these methods have the "salt and pepper" phenomenon due to the large spectral differences in the same object. The object-based methods aim to alleviate this problem by analyzing more holistic information of ground objects [28], [29], [30], [31]. Benedek and Szirányi [30] proposed a conditional mixed Markov model by integrating the local correlation and global intensity statistics. Wang et al. [31] proposed a change detection model for SAR images via a triplet Markov field model by considering the speckle noise and the correlation of nonstationarity in bitemporal SAR images. Although the object-based methods can effectively prevent the "salt and pepper" phenomenon, there are certain difficulties in feature extraction. Generally, the traditional methods always obtain poor results when the changed and unchanged features overlap or the statistical distribution is not accurately modeled, this situation would be worse as the spatial resolution increases.

The DL-based methods have been investigated for several years in change detection of homogeneous remote sensing images and have made remarkable achievements [32]. The existing DL-based change detection methods can be divided into three categories: supervised, semisupervised, and unsupervised. The supervised DL-based change detection methods always have high detection accuracy due to the supervision information. Gong et al. [33] employed a deep neural network to learn the representation of bitemporal SAR images and then obtain the change regions via fine-tuning. Zhang and Lu [34] proposed a spectral–spatial joint learning network for change detection of multispectral images, which considered both spatial and spectral information and fully explored the underlying information of fusion features. Chen et al. [35] integrated the

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siamese convolutional neural network and recurrent neural network into a unified network, which can be employed in the bitemporal homogeneous and heterogeneous images. The semisupervised methods usually train with a small number of samples and then adapt the trained model to the whole dataset. Alvarez et al. [36] proposed self-supervised conditional GAN, which can be directly employed to change detection without any external classifiers and pretrained networks. Zhang et al. [37] first employed CNN to learn the depth features of remote sensing images and then utilized the transfer learning method to construct a dual-channel network with shared weights to detect the difference map. The unsupervised methods need not employ labels as supervision information, which directly obtain the change regions by comparison. Saha et al. [38] employed the supervised semantic segmentation to USCD of multitemporal images. Geng et al. [39] introduced saliency-guided deep neural networks for multitemporal SAR images, which designed a hierarchical fuzzy c-means clustering (HFCM) to automatically select pseudo samples for training.

B. Change Detection for Heterogeneous Remote Sensing Images

Recently, various methods have been proposed to overcome the inconsistent feature representations in heterogeneous images [40], which can be roughly divided into three categories: 1) classification-based methods; 2) DI-based methods; and 3) DL-based methods.

The classification-based methods first conduct individual classifications to obtain the classification maps and then compare the classification maps to generate the change detection result. The post-classification comparison (PCC) [41] method is a classical classification-based change detection method, which is widely used in monitoring land cover used and change analysis [42]. Obviously, there is a problem with error propagation because the performance of change detection strongly relied on classification accuracy. To overcome this problem, compound classification is introduced to classify the current image with the previous image by considering the temporal correlation. For instance, Hedhli et al. [43] presented a joint classification method by considering the correlation of multitemporal images to improve the classification accuracy. Bruzzone et al. [44] employed the joint class probabilities to associate the probability of land cover changes, describing the correlation of multitemporal images more accurately. Subsequently, Wan et al. [45] proposed a change detection method for heterogeneous images by combining compound classification and segmentation to reduce the salt and pepper effect, termed MS-CC. Then, the improved version of MS-CC [46], integrating the cooperative multitemporal segmentation and compound classification, termed CMS-CC, is proposed to improve the detection accuracy. Although classification-based methods are simple and feasible, the classification performance seriously affects detection accuracy, especially when SAR images are difficult to accurately classify due to the speckle noise.

The DI-based methods usually design a method to obtain the DI between heterogeneous images and then employ a threshold or clustering algorithm to determine the change regions. For this method, the key is to design a similarity evaluation operator to obtain the DI. For example, Mercier et al. [47] performed similarity measure by modeling the dependencies of unchanged regions and then obtained the changed regions via Kullback–Leibler (KL) distance. Sun et al. [48] proposed a nonlocal patch similarity-based graph (NPSG) method to represent each image patch via a graph and then determined the changed regions by calculating the consistency between the two graphs of the heterogeneous images. Afterward, an improved NPSG (INPSG) [49] is proposed to boost the robustness and improve the detection accuracy of NPSG, which employed a structure consistency-based graph method to detect the land cover changes in structures of two images. Liu et al. [50] transferred one image space to the other image space via homogeneous pixel transformation and then obtained the changed pixels through Fuzzy-c means clustering algorithm. Sun et al. [51] constructed a graph mapping model (IRGMcS) to describe the structure of heterogeneous images, then compared the graph in the same domain to obtain the DI, and finally detected the changed regions via the Markovian co-segmentation model. In addition, there are also other methods for similarity measure, such as the pixel pair (PP) method [52], the distance of sorted histogram (SH) [53], and the affinity matrix-based image regression (AM-IR) [54]. However, the SM-based methods commonly have challenges in complex scenarios and noisy situations, and it is difficult to measure the similarity of heterogeneous images, which leads to a precipitous degradation in the performance of change detection.

The DL-based methods have developed rapidly in recent years due to their efficient and effective performance. The DL-based methods are divided into supervised methods and unsupervised methods according to whether the labels are required for supervision [55]. For unsupervised methods, Liu et al. [56] proposed a symmetric convolutional coupling network (SCCN) to represent the heterogeneous images in a more consistent feature space, then calculated the DI in the feature space, and finally generated the detection result with a threshold. Liu et al. [16] proposed a USCD model for heterogeneous images, which transformed one image space (e.g., optic image) to another image (e.g., SAR image) for comparison. Luppino et al. [18] proposed two networks (e.g., ACE-Net and X-Net) to alleviate the effect of changing pixels on unsupervised image transformation, which optimized with a prior weighted loss function. Niu et al. [19] translated optic images to the approximated SAR images based on conditional adversarial network (CAN) to reduce the pixel-wise difference for direct comparison. In addition, there are also other unsupervised methods [57], [58], [59] that have been proposed for heterogeneous images, many of which transformed images from one modality to another modality and then performed an unsupervised method based on similarity measure. Obviously, these unsupervised methods overcome the difference between the heterogeneous images through transformation, then calculate the DIs in the same space, and finally obtain the result of change detection through a threshold method. On one hand, the detection accuracy

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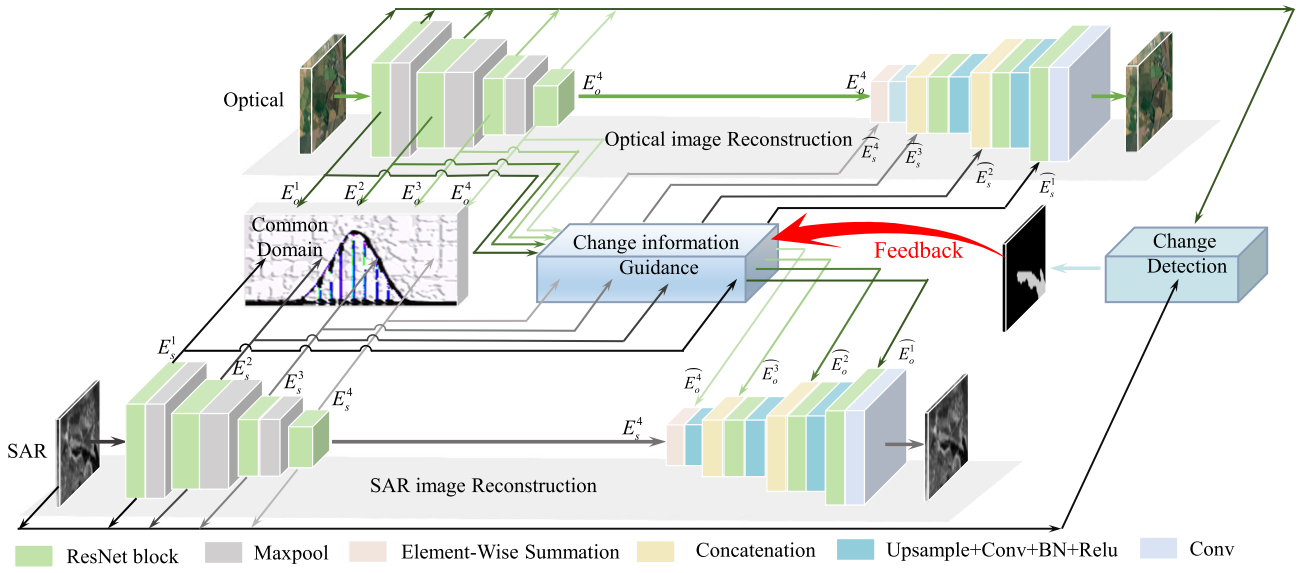


Fig. 1. Overall framework of the proposed DAHCD. Optical and SAR images are initially encoded into multilevel features (e.g., E_o^1, E_o^2, E_o^3, and E_o^4, and E_s^1, E_s^2, E_s^3, and E_s^4), with each level of features being restricted to a common domain. Then, optical features are used to reconstruct SAR image, while SAR features are used to reconstruct the optical image. Simultaneously, the multilevel features and the original SAR and optical image are initially detected for changes. Moreover, the change map is feedback to guide the multilevel features of optical and SAR images for eliminating the influence of changing regions.

is greatly affected by the transformed image, which also has the problem of error propagation. On the other hand, the SM-based unsupervised methods are sensitive to noise and always generate false alarms. For supervised methods, Li et al. [17] proposed a DTCDN, which first performed image transformation through a deep transformation network to reduce the modality difference and then employed a supervised change detection network to determine the change regions. Lv et al. [60] proposed a multiscale UNet (M-UNet) to capture various shapes of land cover types, which learned the abstract features in two image domains via concatenating the heterogeneous images. Although these supervised methods have relatively high detection accuracy, they also have the problem of error propagation (e.g., DTCDN) and cannot fully consider the difference between modalities (e.g., M-UNet).

III. Methodology

In this section, we introduce the proposed change detection of heterogeneous images using a feedback guidance mechanism with the auxiliary of domain adaptive reconstruction, which is mainly divided into the domain adaptive cross reconstruction, change detection network, and feedback guidance mechanism, as shown in Fig. 1. Specifically, we first encode the two images (e.g., optical image and SAR image) into a common domain from the perspective of feature distribution. Then, the features of SAR and optical images are employed to reconstruct the optical and SAR images, respectively. In this dual-adaptive way, the features of optical and SAR images can be translated to an intermediate common domain. Moreover, the transformed multihierarchical features of optical and SAR images and the original heterogeneous optical-SAR image pairs are fed to a change detection network to obtain the changed pixels. In addition, a feedback guidance mechanism is designed to employ the change information to guide the transformed multihierarchical features of optical and SAR images for eliminating the influence of changing regions. Suppose that there are two sensors with different characteristics, which can capture images in the same scenario at different times, denoted as $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ in domain $\mathcal{X}$ (e.g., an optical sensor) and $\mathbf{Y} \in \mathbb{R}^{H\times W\times c}$ in domain $\mathcal{Y}$ (e.g., SAR sensor). $H \times W$ is the size of the spatial dimension of images, and C and c are the bands of optical images and polarizations of SAR images, respectively.

A. Domain Adaptive Cross Reconstruction

Domain adaptation is one of the research tasks in transfer learning, which mainly aims to solve the problem of the inconsistent distribution between the source domain and the target domain [61], [62]. According to different scenarios, domain adaptation can be divided into sample adaptation, feature adaptation, and model adaptation [63]. Recently, there are also some works [64], [65], [66] to employ the idea of domain adaptation to change detection in optical images to alleviate the universal domain shift between the multitemporal optical images. As for change detection of heterogeneous images, this domain shift in the multitemporal and multisource images is more serious due to the different imaging mechanisms. Therefore, in this article, we adopt the idea of domain adaptation to translate heterogeneous images to the same feature domain for comparison. Specifically, we implement domain adaptive reconstruction from two aspects by considering the complementary information and diverse information between different modalities. First, we seek to encode the heterogeneous images (e.g., optical image $\mathbf{X}$ and SAR image $\mathbf{Y}$) into a common space $\mathcal{Z}$ through different encoders to alleviate the huge modalities deference: $\widehat{\mathbf{X}} = E_o(\mathbf{X})$ and $\widehat{\mathbf{Y}} = E_s(\mathbf{Y})$, the encoders $E_o()$ and $E_s()$ aim to transform the optical image $\mathbf{X}$ and the SAR

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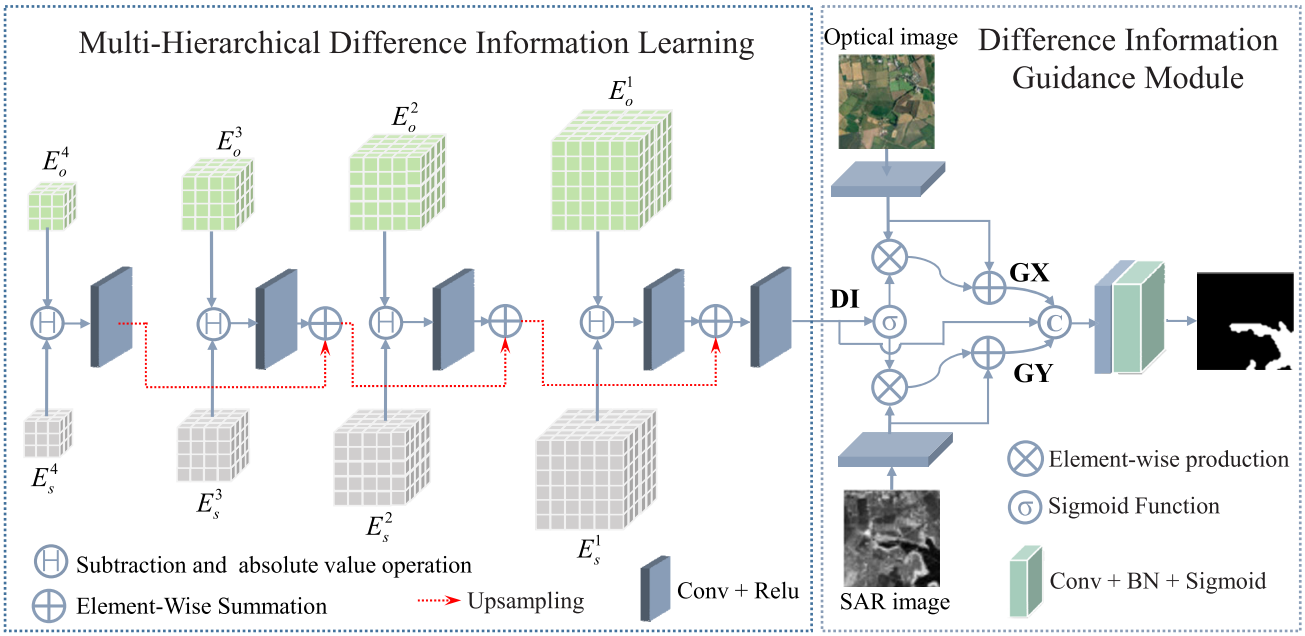


Fig. 2. Framework of the change detection network.

image $\mathbf{Y}$ into a common space $\mathcal{Z}$, respectively. Second, after transformation to the same domain, we assume that one image (e.g., optical image $\mathbf{X}$) can be reconstructed by using the features of another image (e.g., SAR image $\mathbf{Y}$), as described in the following equation:

$$\widetilde{\mathbf{Y}} = D_s(\widehat{\mathbf{X}}) \tag{1}$$

$$\widetilde{\mathbf{X}} = D_o(\widehat{\mathbf{Y}}) \tag{2}$$

where $D_o()$ and $D_s()$ are the decoders to reconstruct the optical image $\widetilde{\mathbf{X}}$ and the SAR image $\widetilde{\mathbf{Y}}$ with the features of the SAR image $\widehat{\mathbf{Y}}$ and the features of the optical image $\widehat{\mathbf{X}}$, respectively.

We employ KL divergence as constraints to optimize the features of each layer of the encoder to maintain the optical and SAR images to the same distribution

$$\mathrm{KL}[P(Z)||Q(Z)] = \sum_{z\in Z}\left[P(z)\log\frac{P(z)}{Q(z)}\right] \tag{3}$$

where P and Q are two asymmetric distributions and Z is a set of random variables. In this way, we can transform optical images in the $\mathcal{X}$ domain and SAR images in the $\mathcal{Y}$ domain to a common domain $\mathcal{Z}$ with small distribution differences.

After aligning the features of optical and SAR images in the common domain $\mathcal{Z}$ via domain adaptation, we can utilize the features of the optical image to reconstruct the SAR image and utilize the features of the SAR image to reconstruct the optical image, respectively. However, it is unfeasible to directly employ the features of an optical image to reconstruct an SAR image or the features of an SAR image to reconstruct an optical image due to the existence of land cover changes between optical and SAR images. To overcome this problem, we creatively propose a feedback guidance mechanism to eliminate the interference of changing regions on image reconstruction, which will be described specifically in Section III-C.

B. Change Detection Network

After domain adaptation, the multiscale homogeneous features of SAR and optical images are obtained, which can be fed to the change detection network to achieve cooperative detection of multiscale change information. Therefore, we design a progressive detection network to detect change regions from small scale to large scale by considering objects with various shapes. In addition, we also take the original optical and SAR images as part of the inputs of the detection network to exploit the diverse information of multimodal images. Fig. 2 shows the framework of the change detection network, we first conduct subtraction followed by an absolute value operation between the homogeneous features at the same hierarchy extracted from SAR and optical images and then exploit the change information at the same hierarchy through convolution operation. Then, we employ the difference information obtained from the upper layer to guide the detection of features on a larger scale. After obtaining the difference information (DI) between multihierarchical homogeneous features of optical and SAR images, a DIGM is proposed to improve the detection accuracy, which employs the DI exploited in the homogeneous features to guide the original heterogeneous images to pay more attention to the information of the changed areas by allocating greater weight to the changed pixel, as expressed in (4)–(6). The proposed DIGM has the ability to guide the detection model to directly adapt to the original heterogeneous images, which can improve the missed detection rate and reduce false positives

$$\mathbf{GX} = \mathrm{sigmoid}(\mathbf{DI}) \odot \mathrm{conv}(\mathbf{X}) + \mathrm{conv}(\mathbf{X}) \tag{4}$$

$$\mathbf{GY} = \mathrm{sigmoid}(\mathbf{DI}) \odot \mathrm{conv}(\mathbf{Y}) + \mathrm{conv}(\mathbf{Y}) \tag{5}$$

$$\mathrm{CDM} = \mathrm{classifier}(\mathrm{cancat}(\mathbf{GX}, \mathbf{GY}, \mathbf{DI})) \tag{6}$$

where sigmoid is the sigmoid function, conv is the convolution operation with a kernel size of 3×3, $\odot$ is the element-wise

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production, cancat is the concatenate operation, and classifier consists of three convolution layers and a batch normalization layer and then follows a sigmoid function for classification. **GX** and **GY** are the guidance features obtained based on **X** and **Y**, respectively, which pay more attention to the pixels in the changing area and can extract the discriminative features for change detection.

C. Feedback Guidance Mechanism

The feedback guidance mechanism is proposed to improve the accuracy of change detection from the following two aspects. The first aspect is to only employ the nonchanged areas of the SAR image (or optical image) to reconstruct the optical image (or SAR image), eliminating the influence of changing regions on the domain adaptive reconstruction. The second aspect is to establish a connection between the two tasks of domain adaptation reconstruction and change detection so that they have the ability to collaborate in optimization and benefit from each other. Specifically, a simple yet effective feedback guidance mechanism is designed to guide the multihierarchical features of optical (e.g., E_o^1, E_o^2, E_o^3, and E_o^4) and SAR (e.g., E_s^1, E_s^2, E_s^3, and E_s^4) images to reconstruct the SAR and optical images, respectively, using the preliminary detection map. In this way, the change information can be employed as a piece of auxiliary information to help the cross reconstruction of SAR and optical images; concurrently, the cross reconstruction of SAR and optical images can be applied as the regularization branch of the change detection network to improve the accuracy. The feedback guidance mechanism is described as follows:

$$\widehat{E_o^1} = \text{repeat}(1 - \text{CDM}) \odot E_o^1 \tag{7}$$

$$\widehat{E_o^2} = \text{repeat}(1 - \text{down}(\text{CDM})) \odot E_o^2 \tag{8}$$

$$\widehat{E_o^3} = \text{repeat}(1 - \text{down}(\text{CDM})) \odot E_o^3 \tag{9}$$

$$\widehat{E_o^4} = \text{repeat}(1 - \text{down}(\text{CDM})) \odot E_o^4 \tag{10}$$

$$\widehat{E_s^1} = \text{repeat}(1 - \text{CDM}) \odot E_s^1 \tag{11}$$

$$\widehat{E_s^2} = \text{repeat}(1 - \text{down}(\text{CDM})) \odot E_s^2 \tag{12}$$

$$\widehat{E_s^3} = \text{repeat}(1 - \text{down}(\text{CDM})) \odot E_s^3 \tag{13}$$

$$\widehat{E_s^4} = \text{repeat}(1 - \text{down}(\text{CDM})) \odot E_s^4 \tag{14}$$

where CDM is the change detection map, repeat() is the duplicated operation to duplicate the matrix in the channel dimension to make the two matrixes in the same size for pixel-wise multiplication, and down() is the downsampled operation with the bicubic interpolation algorithm.

D. Loss Functions

In this article, we design a weight decay strategy to train the proposed model because the domain adaptive reconstruction and change detection are collaboratively optimized via a feedback guidance mechanism. In the early stage of network training, we should assign a higher weight to the domain adaptive reconstruction so that they can strongly constrain the network to reconstruct better SAR and optical images and guide the network to encode the optical and SAR images into a common domain. With the progress of training, since the whole network should pay more attention to the change detection branches, we gradually reduce the weight of domain adaptive reconstruction tasks and transform the focus of network training to the task of change detection. In this way, the network can adaptively focus on the different branches with the progress of the training phases to improve the network performance. The total loss consists of change detection loss (e.g., $\mathcal{L}_{\text{cd}}$), optical and SAR image reconstruction loss (e.g., $\mathcal{L}_{\text{optical}}$ and $\mathcal{L}_{\text{SAR}}$), and domain adaptive loss (e.g., $\mathcal{L}_{\text{da}}$), which is computed as follows:

$$\mathcal{L}_{\text{cd}} = \mathcal{L}_{\text{focal}}(\text{CDM}, \text{GT}) + \mathcal{L}_{\text{dice}}(\text{CDM}, \text{GT}) \tag{15}$$

where CDM denotes the change detection map and GT is the ground truth of the change detection

$$\mathcal{L}_{\text{focal}}(p) = -\alpha(1-p)^{\gamma}\log(p) \tag{16}$$

where p represents the probability of equivalence between CDM and GT, α is the weighting factor, and γ is focusing parameter to adjust the weights assigned to the harder-to-learn examples

$$\mathcal{L}_{\text{dice}}(\text{CDM}, \text{GT}) = 1 - \frac{1}{m}\sum_{j=1}^{m}\frac{2\sum_{i=1}^{N}\text{CDM}_{i,j}\text{GT}_{i,j}}{\sum_{i=1}^{N}\text{CDM}_{i,j} + \sum_{i=1}^{N}\text{GT}_{i,j}} \tag{17}$$

where CDM_i, j and GT_i, j represents the value of the CDM and GT at point (i, j), respectively

$$\mathcal{L}_{\text{SAR}} = 1 - \text{SSIM}(\widehat{\mathbf{Y}}, \mathbf{Y}) \tag{18}$$

$$\mathcal{L}_{\text{optical}} = 1 - \text{SSIM}(\widehat{\mathbf{X}}, \mathbf{X}) \tag{19}$$

$$\text{SSIM}(a, b) = \frac{2u_a u_b + C_1}{u_a^2 + u_b^2 + C_1} \cdot \frac{2\sigma_{[a,b]} + C_2}{\sigma_a^2 + \sigma_b^2 + C_2} \tag{20}$$

$$\mathcal{L}_{\text{da}} = \text{KL}\left(E_s^1, E_o^1\right) + \text{KL}(E_s^2, E_o^2) + \text{KL}\left(E_s^3, E_o^3\right) + \text{KL}\left(E_s^4, E_o^4\right) \tag{21}$$

where u_a and u_b are the mean values of a and b, respectively, σ_a and σ_b are the standard deviation of a and b, respectively, and C_1 and C_2 are constant to prevent the denominator from being zero. The total loss function is defined as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cd}} + \frac{N-n}{N}\mathcal{L}_{\text{SAR}} + \frac{N-n}{N}\mathcal{L}_{\text{optical}} + \frac{N-n}{N}\mathcal{L}_{\text{da}} \tag{22}$$

where N is the total epoch and n is the current epoch at the training stage.

IV. Experiments

In this section, we conduct experiments on four publicly available benchmark datasets to comprehensively verify the proposed DAHCD.

A. Datasets and Evaluation Criteria

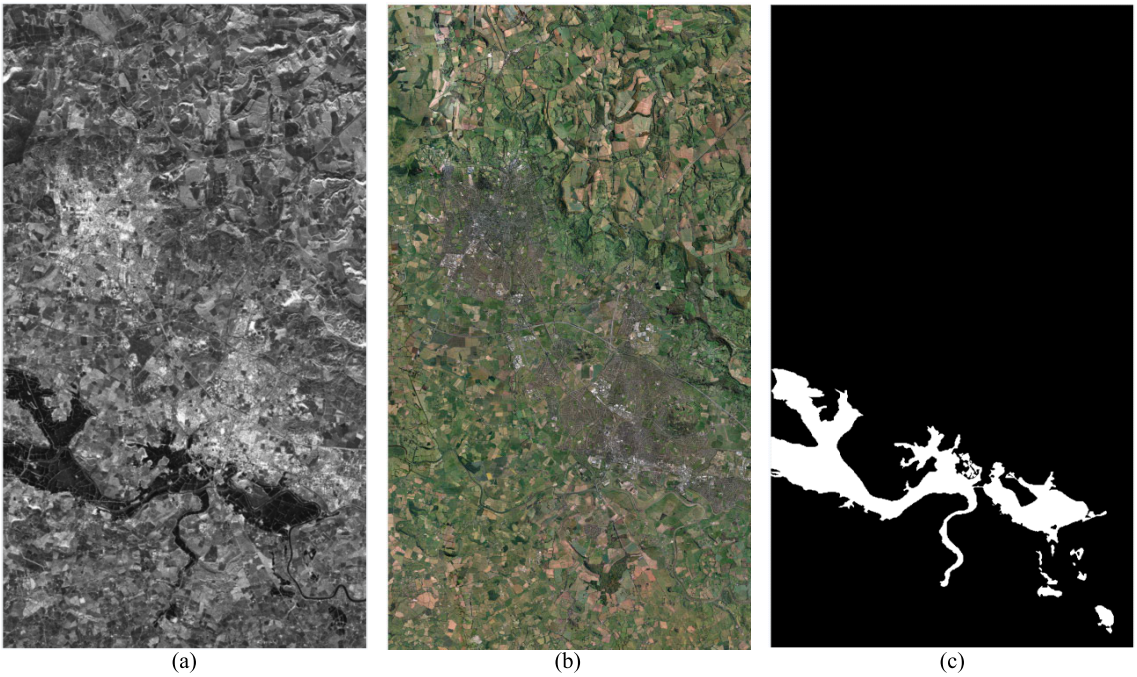
The first dataset is acquired at ShuGuang village in Dongying (China) [67], termed ShuGuang, which consists of two images (i.e., the SAR image with size 921 × 593 × 1 and the optical image with size 921 × 593 × 3) and the ground truth

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(a) (b) (c)

Fig. 3. ShuGuang dataset. (a) SAR image. (b) Optical image. (c) Ground truth.



(a) (b) (c)

Fig. 4. Gloucester I dataset. (a) SAR image. (b) Optical image. (c) Ground truth.

with a size of 921 × 593 × 1. The SAR image and the optical image were captured by Radarsat-8 on June 2008 and Google Earth in 2012, respectively, as shown in Fig. 3(a) and (b). The ground truth illustrated the changes from farmlands to buildings and the river changes in the bottom right, as shown in Fig. 3(c).

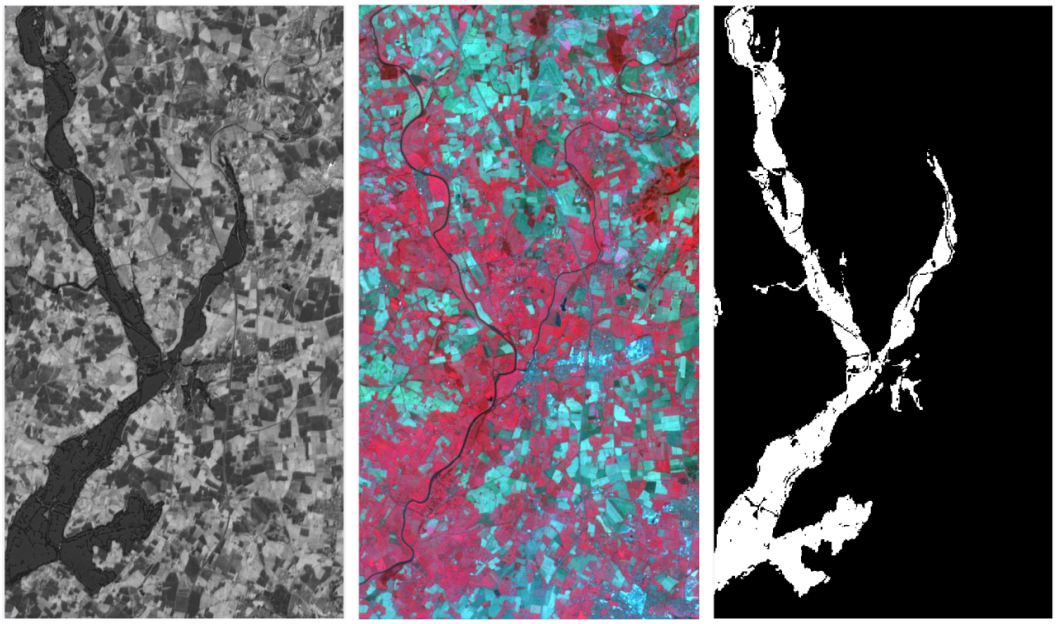
The second dataset is acquired at Gloucester (U.K.) [67], termed Gloucester I, which consists of two images (i.e., the SAR image with size 2325 × 4135 × 1 and the optical image with size 2325 × 4135 × 3) and the ground truth with the size of 2325 × 4135 × 1. The SAR image and the optical image were captured by TerraSAR-X in July 2007 and Quickbird-2 in July 2006, respectively, as shown in Fig. 4(a) and (b). The ground truth showed the land cover changes after a flood, as shown in Fig. 4(c).

The third dataset is also acquired at Gloucester (U.K.) [68], termed Gloucester II, which consists of two images (i.e., the SAR image with size 554 × 990 × 1 and the optical image with size 554 × 990 × 3) and the ground truth with the size of 554 × 990 × 1. The SAR image and the optical image were captured by ERS-1 on November 2000 and SPOT on September 1999, respectively, as shown in Fig. 5(a) and (b). The ground truth also illustrated the land cover changes after a flood, as shown in Fig. 5(c).

The fourth dataset is acquired at California (USA) [54], which consists of two images (i.e., the SAR image with size 2000 × 3500 × 3 and the optical image with size 2000 × 3500 × 11) and the ground truth with the size of 2000 × 3500 × 1. The SAR image was captured by Sentinel 1-A on 18 February 2017, which consists of three bands recorded in the polarization directions of VV and VH, and the third channel depicts the intensity ratio between these two channels, as shown in Fig. 6(a). The optical image was captured by Landsat 8 on January 2017, as shown in Fig. 6(b). The ground truth showed the land cover changes after a flood, as shown in Fig. 6(c).

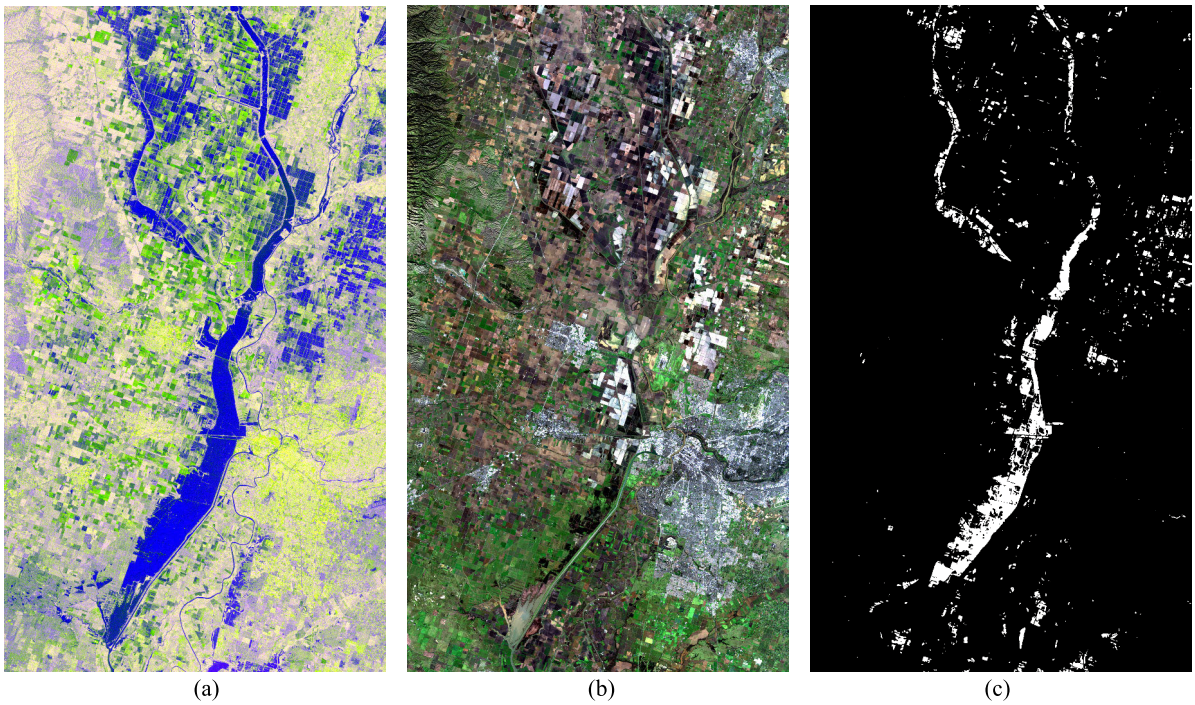
To measure the performance of our proposed DAHCD, several frequently used metrics are employed, including precision (P), recall (R), F1-score (F1), overall accuracy (OA), and Kappa coefficient (KC). F1-score comprehensively considered the false alarm rate and missed alarm rate, which accurately reflects the accuracy of change detection relatively. OA reflects the accuracy of the change detection results, but it is insufferable for OA when the number of samples is unbalanced. KC is

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(a) (b) (c)

Fig. 5. Gloucester II dataset. (a) SAR image. (b) Optical image. (c) Ground truth.



(a) (b) (c)

Fig. 6. California dataset. (a) SAR image. (b) Optical image. (c) Ground truth.

widely used as an indicator of consistency to compensate for the deficiency of OA.

B. Comparison Methods and Implementation Details

To verify the effectiveness of our proposed DAHCD, we compared several representative and effective methods (e.g., PCC [41], IGR-McS [51], CAN [19], SCCN [56], ACE-Net [18], X-Net [18], M-UNet [60], and DTCDN [17]) for heterogeneous images change detection. It is noteworthy that these comparison methods are reimplemented under fair parameter settings and training data with our proposed DAHCD. PCC and IGR-McS are unsupervised traditional methods, and PCC first classified the heterogeneous images into several categories and then identified the change regions by comparing each category independently. IGR-McS first calculated the DI in a common domain through a graph mapping model and then detected the changed regions via the Markovian co-segmentation model. Both two methods have the problem of error propagation. CAN, SCCN, ACE-Net, and X-Net are the unsupervised DL-based methods, and CAN

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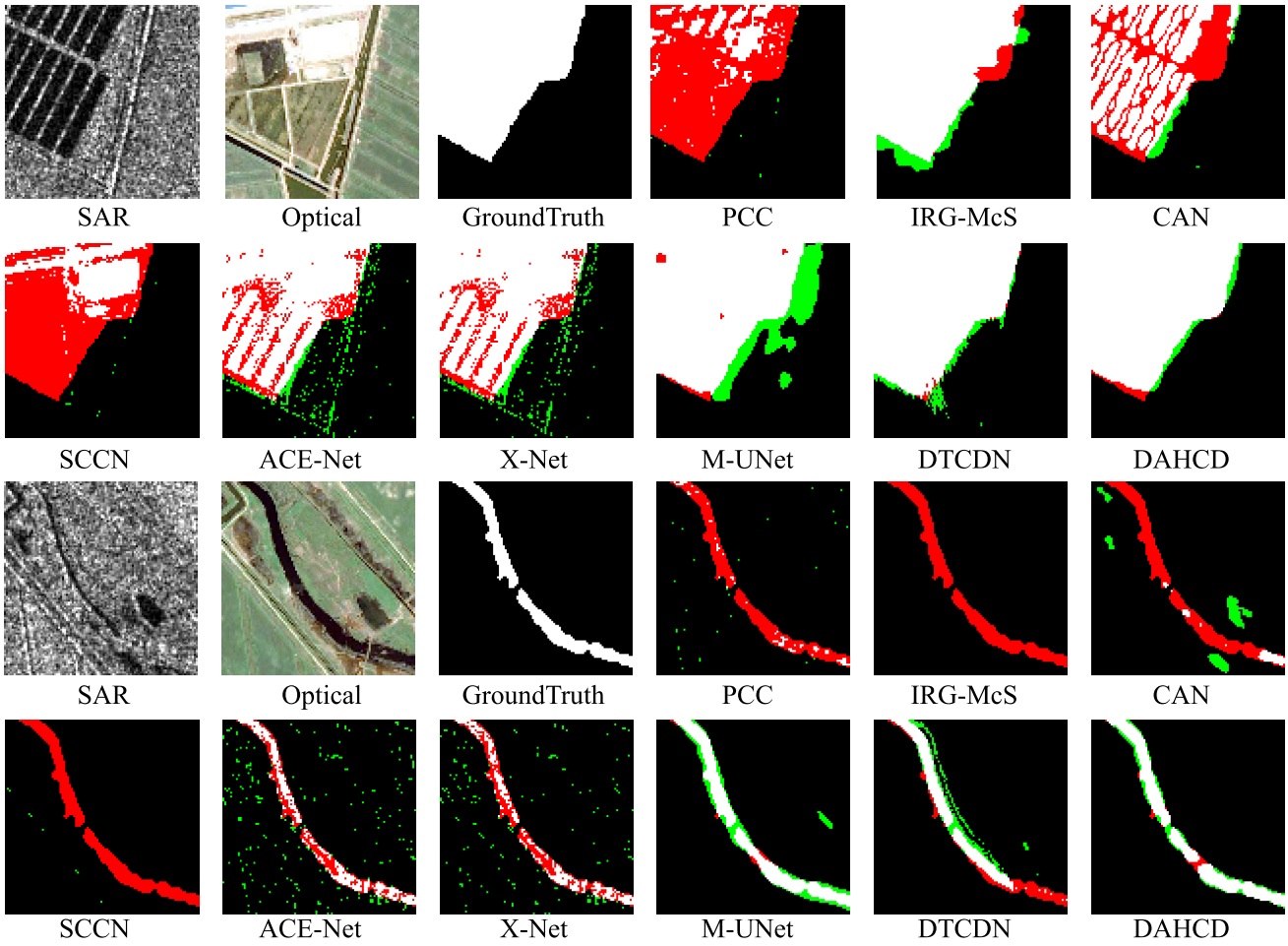


Fig. 7. Qualitative results of different change detection methods on the ShuGuang dataset. White is used to represent true positives (TP), while black represents true negatives (TN). Red signifies false positives (FP) and green indicates false negatives (FN).

overcomes the difference between heterogeneous images using conditional GAN (cGAN) to translate optic images to the approximated SAR images and then obtains the changed regions by direct comparison. SCCN calculated the DI in a consistent feature space and then obtained the result of change detection via a threshold algorithm. ACE-Net and X-Net also first transform the heterogeneous images into a common feature space and then employ a threshold algorithm to the calculated DI to obtain the result of change detection. These unsupervised DL-based methods are not only sensitive to noise and always generate false alarms but also have the problem of error propagation. M-UNet and DTCDN are the supervised DL-based methods, and M-UNet directly concatenated the heterogeneous images and then fed them into the proposed multiscale U-Net by considering the multiscale information, which does not consider the imaging differences of heterogeneous images. DTCDN first employs an image-to-image transformation network and then designs a depth-wise separable convolution-based U-Net++ for change detection. Obviously, the DTCDN also has the problem of error propagation and cannot exploit the complementary information in heterogeneous images. Different from the above methods, our proposed DAHCD integrates the domain adaptation and change detection into a unified network, domain adaptation positively improves the performance of change detection, and the result of change detection reversely guides the domain adaptive cross reconstruction.

In the image reconstruction network of our proposed DAHCD, a U-Net-like network and the ResNet block were employed as the feature extraction unit, and the number of convolutional filters is set to 64, 128, 256, and 512 in the four encoder layers. In the experiments, the default values of the compared methods are set; for our proposed DAHCD, the Adam optimization is employed and the initial learning rate of 0.001 is set with exponential decay. The batch size and the epochs are set to 8 and 200, respectively. The PCC and IGR-McS are implemented in python 3.7 and MATLAB R2020b, respectively, on 64-GB Intel Xeon Gold 5218. The DL-based methods are performed on GPU NVIDIA P5000.

C. Experiments on the ShuGuang Dataset

The ShuGuang dataset mainly contains the land cover changes from farmlands to buildings and the river changes. Considering the limited size of ShuGuang, we employed cross validation in our experiments to ensure comprehensive coverage across all regions of the image. In each validation phase, we selected 80% and 20% of the entire image for training and testing, respectively. In this way, we need to divide the whole

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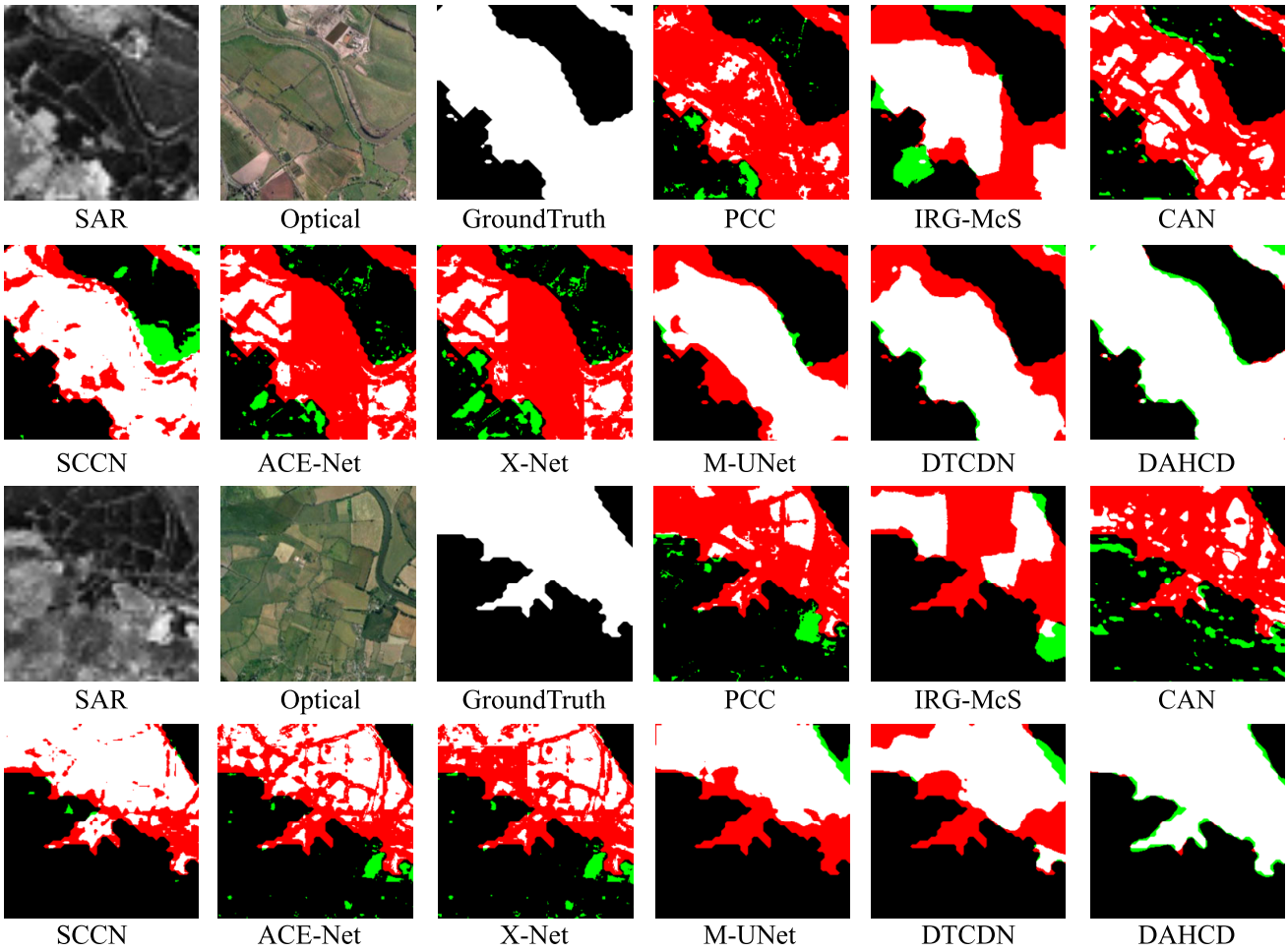


Fig. 8. Qualitative results of different change detection methods on the Gloucester I dataset. White is used to represent true positives (TP), while black represents true negatives (TN). Red signifies false positives (FP) and green indicates false negatives (FN).

dataset five times to make the test set cover the entire image. The visual results of different change detection methods are shown in Fig. 7. In the results, the red area represents the instances of missed detection, while the green area indicates the occurrences of false detection. We can find that the PCC can only detect the limited change regions and show the worst performance of all methods. IRGMcS shows a better detection performance than PCC and can detect more change regions. As for DL-based methods, the unsupervised methods, including CAN, SCCN, ACE-Net, and X-Net, show more advantages than traditional methods, especially compared with PCC. The supervised methods (e.g., M-UNet and DTCDN) exhibit the superior performance than the compared traditional methods and unsupervised DL-based methods. Our proposed DAHCD shows more advantageous detection results, which have a lower rate of false alarms and have the ability to detect more changing regions.

In quantitative evaluation, we use the average of five validation phases as the final evaluation result of the entire image. The quantitative evaluation results are shown in Table I, and the evaluation results show similar conclusions to the qualitative results. As can be seen, our proposed DAHCD achieves superior performance than the compared methods in all quantitative metrics.

TABLE I

QUANTITATIVE EVALUATION RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON THE SHUGUANG DATASET

Methods	Precision	Recall	F1	OA	KC
PCC	0.6455	0.1909	0.2946	0.9580	0.2795
IGR-McS	0.8332	0.7742	0.8026	0.9825	0.7935
CAN	0.3472	0.3980	0.3709	0.9379	0.3384
SCCN	0.8414	0.3664	0.5101	0.9677	0.4965
ACE-Net	0.4681	0.7806	0.5853	0.9492	0.5600
X-Net	0.5239	0.7880	0.6294	0.9573	0.6077
M-UNet	0.7762	0.9106	0.8508	0.9742	0.8503
DTCDN	0.7291	0.8681	0.8017	0.9703	0.7917
DAHCD	**0.8407**	**0.9194**	**0.8809**	**0.9811**	**0.8709**

D. Experiments on the Gloucester I Dataset

Gloucester I dataset mainly embraces the land cover changes after a flood disaster, and the whole image is cropped into image patches with the size of 256 × 256. We randomly selected 60% of the whole data for training and the rest 40% for testing, and two example results of different change detection methods are shown in Fig. 8. In the results, the red area represents instances of missed detection, while the green area indicates occurrences of false detection. Obviously, our proposed DAHCD shows the best performance. PCC performs the worst results of all methods. IGR-McS shows

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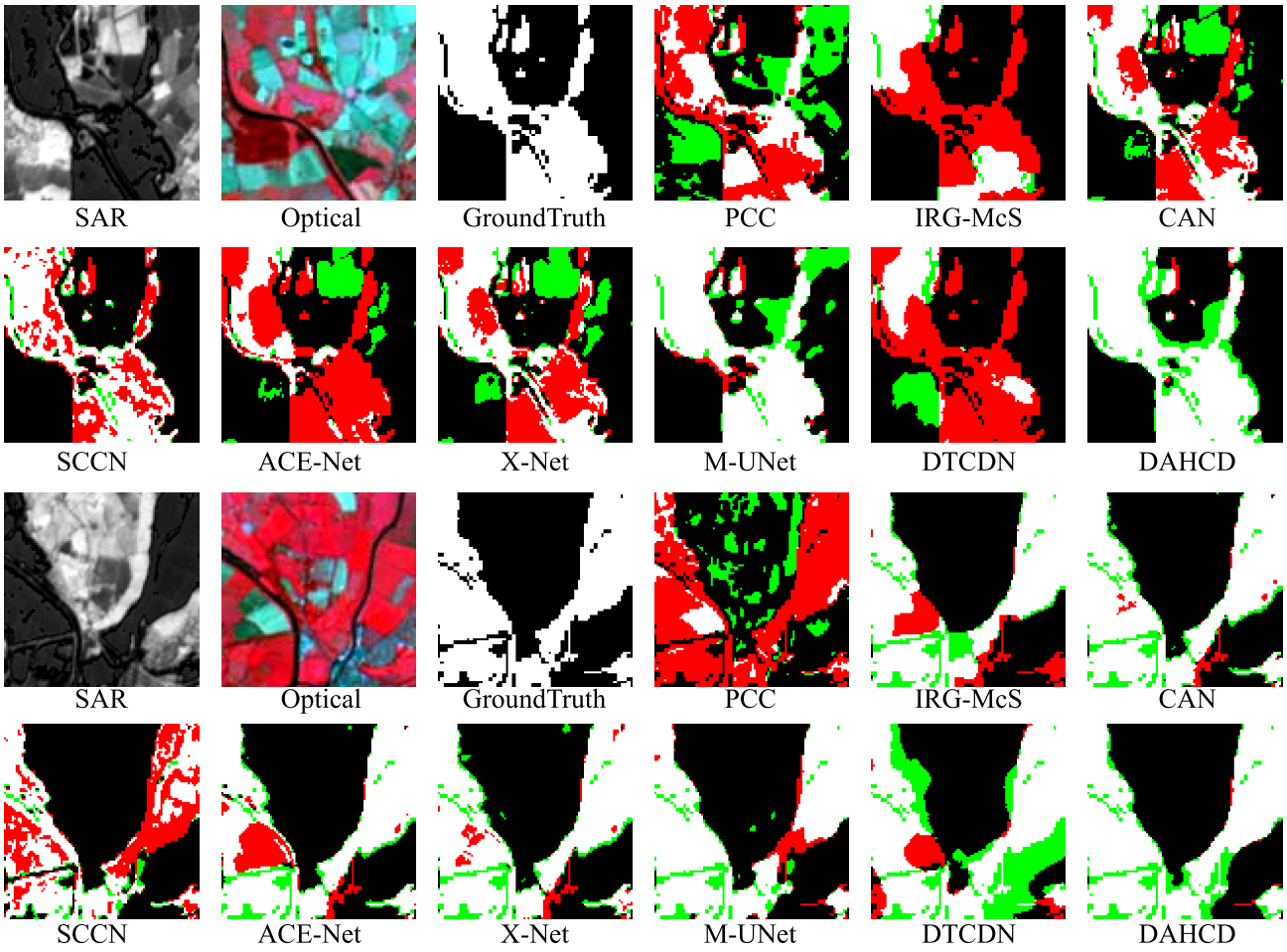


Fig. 9. Qualitative results of different change detection methods on the Gloucester II dataset. White is used to represent true positives (TP), while black represents true negatives (TN). Red signifies false positives (FP) and green indicates false negatives (FN).

TABLE II
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON THE GLOUCESTER I DATASET

Methods	Precision	Recall	F1	OA	KC
PCC	0.1765	0.2389	0.2030	0.8796	0.1395
IGR-McS	0.6018	0.6669	0.6327	0.9503	0.6061
CAN	0.4153	0.4068	0.4110	0.9252	0.3711
SCCN	0.6710	0.8075	0.7329	0.9622	0.7128
ACE-Net	0.3729	0.4926	0.4245	0.9126	0.3782
X-Net	0.2885	0.3753	0.3263	0.8986	0.2725
M-UNet	**0.9490**	0.6520	0.7717	0.9738	0.7583
DTCDN	0.7941	0.7203	0.7559	0.9683	0.7390
DAHCD	0.8777	**0.8342**	**0.8554**	**0.9808**	**0.8451**

acceptable detection results, but it still misses many changed pixels. The unsupervised DL-based methods, including CAN, SCCN, ACE-Net, and X-Net, show more advantages than traditional methods, but there are numerous discontinuous noises in their results due to the threshold-based classification method. The supervised DL-based methods (e.g., M-UNet and DTCDN) show superior performance, but the results still miss many changed pixels. Compared with the above methods, our proposed DAHCD exhibits the best detection performance.

The quantitative evaluation results are shown in Table II, and the evaluation results show similar conclusions to the qualitative results. As can be seen, our proposed DAHCD achieves superior performance than the compared methods, which achieve the best scores in recall, F1-score, OA, and KC but also achieve the second-best score in precision.

E. Experiments on the Gloucester II Dataset

Gloucester II dataset also mainly embraces the land cover changes after a flood disaster, and the whole image is cropped into image patches with the size of 256 × 256. To ensure fairness in comparison, we adopt the same validation strategy as the ShuGuang dataset due to the limited size of this dataset. The results of different change detection methods are shown in Fig. 9. In the results, the red area represents instances of missed detection, while the green area indicates the occurrences of false detection. As can be seen, PCC is basically unable to effectively detect the flood regions, showing the worst performance. IGR-McS can basically detect the flood area, but many change pixels are lost. The unsupervised DL-based methods (e.g., CAN, SCCN, ACE-Net, and X-Net) can also detect the flood region, but still lost many change pixels and introduced some false positives. M-UNet can effectively detect the flood region but introduce several false positives in the result. DTCDN has lost many

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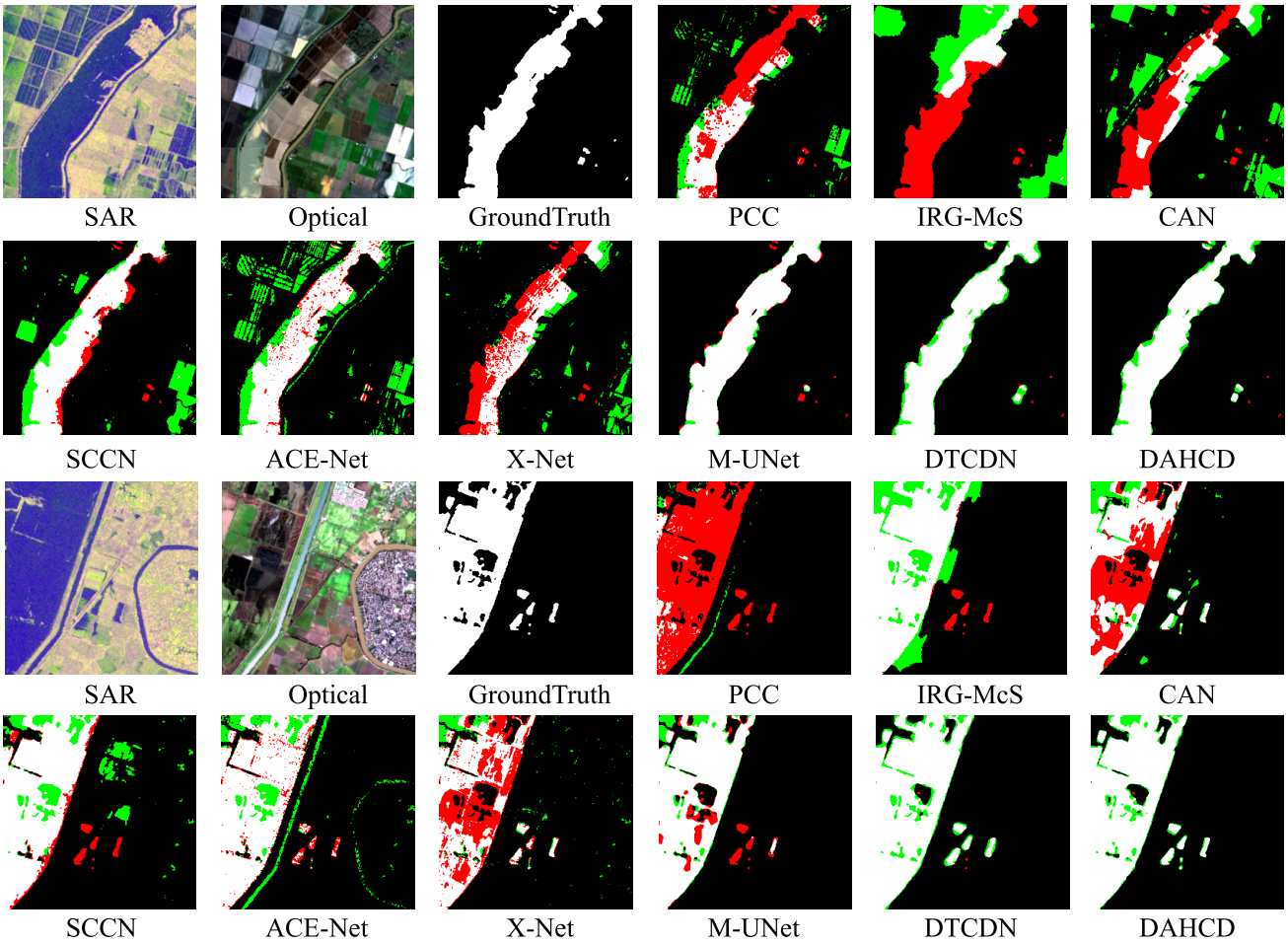


Fig. 10. Qualitative results of different change detection methods on the California dataset. White is used to represent true positives (TP), while black represents true negatives (TN). Red signifies false positives (FP) and green indicates false negatives (FN).

change regions and introduced false alarm pixels due to the influence of image transformation. Our proposed DAHCD can effectively show the whole flood area, although a small number of false positives are introduced. Generally, our proposed DAHCD achieves the best performance.

In quantitative evaluation, we use the average of five validation phases as the final evaluation result of the entire image. The quantitative evaluation results are shown in Table III, which also demonstrates that our proposed DAHCD achieves the best performance. As seen, the proposed DAHCD achieves the best scores in recall, F1-score, OA, and KC but also achieves the second-best score in precision.

TABLE III
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON THE GLOUCESTER II DATASET

Methods	Precision	Recall	F1	OA	KC
PCC	0.2856	0.6088	0.3888	0.7717	0.2703
IGR-McS	0.7915	0.8383	0.8142	0.9544	0.7882
CAN	0.6652	0.8583	0.7495	0.9316	0.7106
SCCN	**0.9379**	0.7650	0.8427	0.9659	0.8238
ACE-Net	0.6276	0.7756	0.6938	0.9183	0.6473
X-Net	0.6076	0.8336	0.7028	0.9159	0.6553
M-UNet	0.6780	0.9508	0.7637	0.9279	0.7428
DTCDN	0.6691	0.8291	0.7391	0.9108	0.6971
DAHCD	0.8285	**0.9776**	**0.8795**	**0.9706**	**0.8659**

F. Experiments on the California Dataset

The California dataset also mainly embraces the land cover changes after a flood disaster, and the whole image is cropped into image patches with the size of 256 × 256. We randomly selected 60% of the whole data for training and the rest 40% for testing, and two example results of different change detection methods are shown in Fig. 10. In the results, the red area represents the instances of missed detection, while the green area indicates the occurrences of false detection. As seen, PCC still obtains the worst performance. IGR-McS shows limited detection performance in two scenarios, which cannot effectively detect the change regions in the first scenario and has numerous false alarm pixels in the second scenario. The unsupervised DL-based methods (e.g., CAN, SCCN, ACE-Net, and X-Net) can roughly detect the change regions, but there are still lost many change pixels and introduced some false positives. M-UNet can effectively detect the change regions but lost many change pixels. DTCDN can also effectively detect the change regions but introduce several false positives in the result. The proposed DAHCD achieves the best performance, which has fewer false alarm pixels.

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TABLE IV

QUANTITATIVE EVALUATION RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON THE CALIFORNIA DATASET

Methods	Precision	Recall	F1	OA	KC
PCC	0.3342	0.2266	0.2701	0.9465	0.2434
IGR-McS	0.4018	0.6353	0.4923	0.9427	0.4636
CAN	0.3106	0.4868	0.3792	0.9304	0.3442
SCCN	0.3558	0.5775	0.4403	0.9358	0.4083
ACE-Net	0.3742	0.6984	0.4873	0.9358	0.4563
X-Net	0.4040	0.4768	0.4374	0.9464	0.4094
M-UNet	**0.7754**	0.7881	0.7817	0.9793	0.7708
DTCDN	0.6174	0.8306	0.7083	0.9679	0.6917
DAHCD	0.7622	**0.8915**	**0.8218**	**0.9818**	**0.8123**

TABLE V

ABLATION EXPERIMENTS FOR DOMAIN ADAPTATION ON THE CALIFORNIA DATASET

Methods	Precision	Recall	F1	OA	KC
w/o FA	0.7450	0.8607	0.7991	0.9788	0.7867
w/o CR	0.6459	0.8699	0.7441	0.9708	0.7281
w/o DA	0.6126	0.8592	0.7102	0.9649	0.6947
DAHCD	**0.7622**	**0.8915**	**0.8218**	**0.9818**	**0.8123**

The quantitative evaluation results are shown in Table IV, which also demonstrates that our proposed DAHCD achieves the best performance. As seen, the proposed DAHCD achieves the best scores in recall, F1-score, OA, and KC but also achieves the second-best score in precision.

V. DISCUSSION

In this section, the effectiveness of our DAHCD is further discussed. The effects of the domain adaptation (e.g., the feature adaptation and cross reconstruction) and the feedback guidance mechanism are analyzed. In addition, the effects of the main structures of the change detection network and the loss functions are also discussed. The experiments are conducted on the California dataset due to the large size of the data and the challenging characteristics.

A. Effect of Domain Adaptation

We first conduct ablation experiments to evaluate the effectiveness of domain adaptation from two aspects (e.g., feature adaptation and cross reconstruction). For feature adaptation, we directly remove the way of distribution alignment and only employ the cross reconstruction for domain adaptation, termed w/o FA. As for cross reconstruction, we remove the way of cross reconstruction and only employ the feature adaptation for domain adaptation, termed w/o CR. Moreover, we also remove the feature adaptation and cross reconstruction to evaluate the effectiveness of domain adaptation, termed w/o DA. The quantitative evaluation results are shown in Table V. As seen, the feature adaptation and the cross reconstruction have different effects on detection performance, among which cross reconstruction has a greater influence on the model detection performance.

B. Effect of Feedback Guidance Mechanism

The feedback guidance mechanism can eliminate the influence of changing regions on cross reconstruction to a certain extent, and we directly remove the feedback guidance mechanism to evaluate its effectiveness, termed w/o FGM. The quantitative evaluation results are shown in Table VI. We can find that the feedback guidance mechanism influences the model performance and the performance of the cross reconstruction is affected by the pixels in the changed regions, which leads to the poor performance of domain adaptation.

TABLE VI

ABLATION EXPERIMENTS FOR FEEDBACK GUIDANCE MECHANISM ON THE CALIFORNIA DATASET

Methods	Precision	Recall	F1	OA	KC
w/o FGM	0.6682	0.8627	0.7488	0.9731	0.7298
DAHCD	**0.7622**	**0.8915**	**0.8218**	**0.9818**	**0.8123**

TABLE VII

ABLATION EXPERIMENTS FOR CHANGE DETECTION NETWORK ON THE CALIFORNIA DATASET

Methods	Precision	Recall	F1	OA	KC
w/o DIGM	0.7128	0.8792	0.7824	0.9767	0.7905
DAHCD	**0.7622**	**0.8915**	**0.8218**	**0.9818**	**0.8123**

TABLE VIII

ABLATION EXPERIMENTS FOR CHANGE DETECTION NETWORK ON THE CALIFORNIA DATASET

Methods	Precision	Recall	F1	OA	KC
L1 loss	0.6753	0.8900	0.7679	0.9747	0.7548
L2 loss	0.6921	0.8869	0.7775	0.9761	0.7651
w/o WDS	0.6825	0.8848	0.7692	0.9757	0.7602
DAHCD	**0.7622**	**0.8915**	**0.8218**	**0.9818**	**0.8123**

C. Effect of Change Detection Network

In the change detection network, a DIGM is proposed by considering the original heterogeneous images. To evaluate the effectiveness of the proposed DIGM, we remove the DIGM and directly feed the DI exploited in the homogeneous features into the classifier to determine the change regions, termed w/o DIGM. The quantitative evaluation results are shown in Table VII. We can find that the original SAR and optical images also play important roles in the change detection of heterogeneous images, and only employing the transformed features inevitably leads to detection errors.

D. Effect of Loss Functions

In this article, we design a weight decay strategy to train the proposed model because the domain adaptive reconstruction and change detection are collaboratively optimized via a feedback guidance mechanism. To evaluate the effectiveness of the weight decay strategy in the loss functions, we remove the weight decay strategy and directly add all loss functions, termed w/o WDS. In addition, we replace the loss function for reconstruction task with L1 and L2 to illustrate the advantages of utilizing 1-SSIM as the loss of reconstruction task. The quantitative evaluation results are shown in Table VIII. We can find that the weight decay strategy and the loss of reconstruction task also influence the performance of the model.

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The network can adaptively adjust the optimization objectives with the proposed weight decay strategy, thereby improving the performance of the model.

VI. Conclusion

In this article, we propose domain adaptation-based change detection for heterogeneous remote sensing images, termed DAHCD. The basic idea is to encode the heterogeneous remote sensing images (e.g., optical and SAR images) into an intermediate common domain to reduce modality differences. To achieve this, we adopt the idea of domain adaptation to the change detection of heterogeneous remote sensing images. Specifically, we design a cooperative domain adaptation from both feature adaptation and cross reconstruction to map the heterogeneous remote sensing images into an intermediate common domain. Feature adaptation alleviates the gap between domains in the way of distribution alignment by constraining the distribution of multihierarchical features. Cross reconstruction employs the features of the optical image to reconstruct the SAR image and employs the features of the SAR image to reconstruct optical images. From the two perspectives of feature adaptation and cross reconstruction, the features of heterogeneous images can be adaptively encoded into the intermediate common domain. Afterward, the change detection network is designed to consider the transformed multihierarchical homogeneous features of optical and SAR images and the original heterogeneous optical-SAR image pairs to identify the changed regions. Moreover, a feedback guidance mechanism is proposed to eliminate the influence of changing regions on the domain adaptive cross reconstruction by utilizing the result of change detection to guide the cross reconstruction. In addition, a hybrid loss function with a weight decay strategy is proposed to optimize the proposed DAHCD to improve its performance. Compared with other methods, our proposed DAHCD can obtain the best scores in recall, F1-score, OA, and KC but also achieve the second-best score in precision.

In our future work, we will devote ourselves to investigating the change detection methods with higher precision for more challenging remote sensing image pairs (e.g., the image pairs with different spatial resolution, the image pairs with very high spatial resolution, and the image pairs with long time series). In addition, we will focus on the tradeoff between computational complexity and model accuracy.

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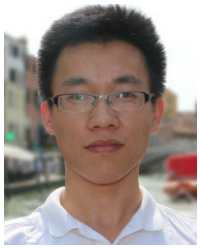
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